

Acute Bilateral Leg Edema Provoked by Spinal Stenosis with CRPS Mechanism: A Case Report

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PCI-IOSS Presentation

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Case Presentation



51-year-old female

Symptoms:

- Chronic of lower back pain accompanied by tingling sensations in both legs
- Bilateral limb edema with skin colour changes

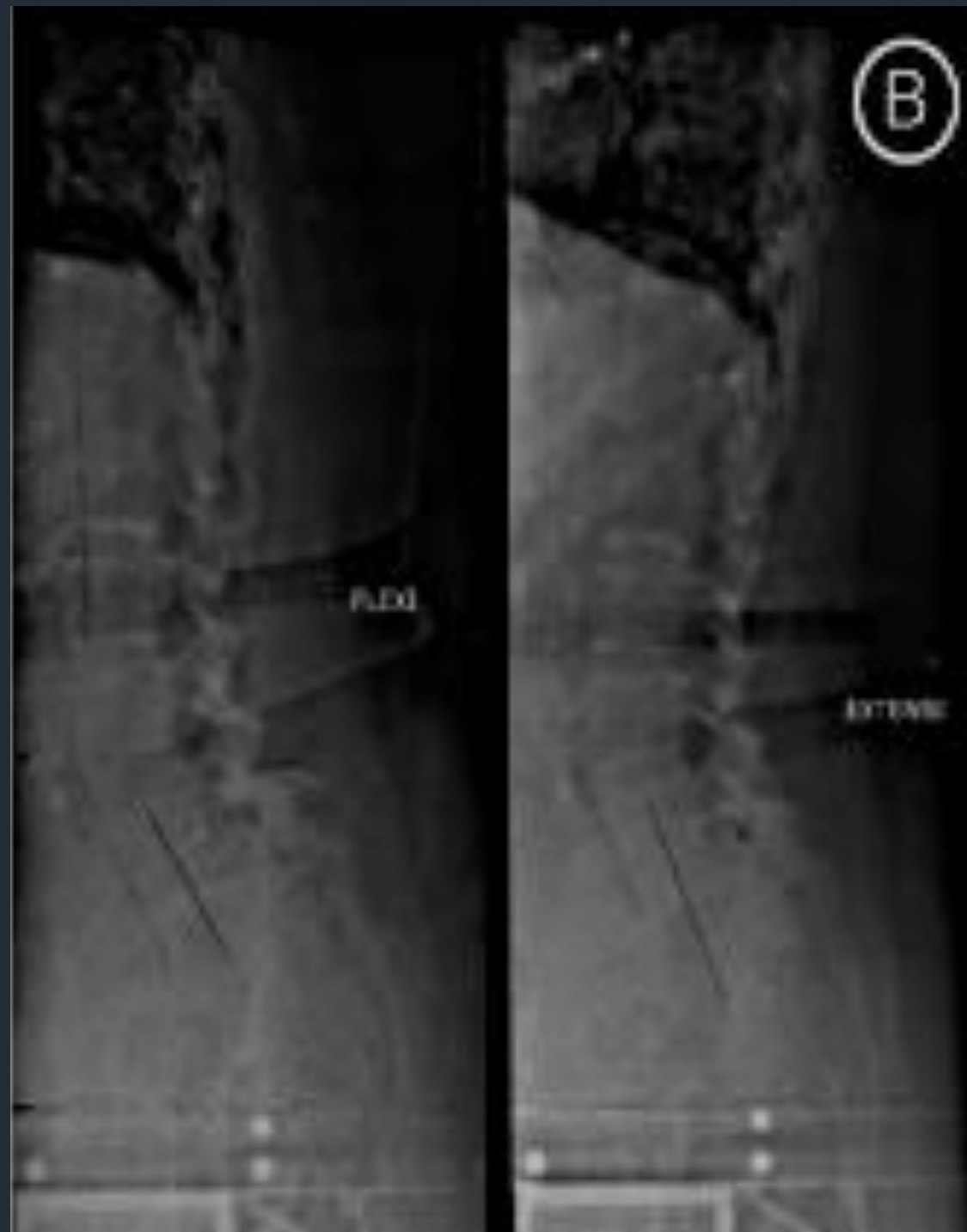
Medical History:

- No history of trauma
- Anti-inflammatory drugs (NSAIDs)
- Minimal relief from previous physiotherapy

Examinations:

- Cardiovascular and metabolic exams were normal
- Neurological status within normal limits
- Body Mass Index : 26 (Overweight)

Case Presentation



Bilateral Edema on Lower Leg (A), Lateral Lumbar X-ray (B), Lumbar MRI (C)

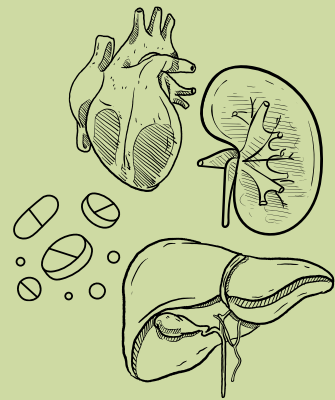
Case Presentation



X-ray of the lumbosacral postoperative (A), Edema on the bilateral limb was resolved postoperative (B)

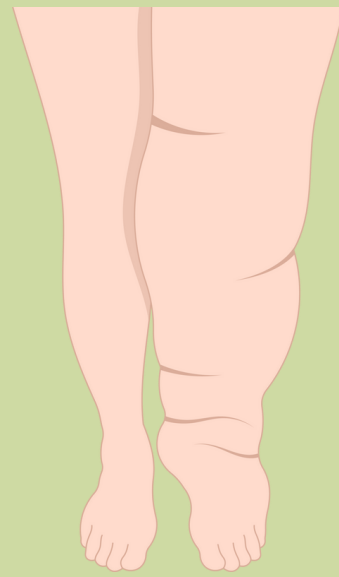
Follow-up: 2 months later, patient resumed normal daily activities

Introduction



- **Bilateral leg edema**
Can result from various diseases, including complex regional pain syndrome could be provoked by spinal stenosis
- **Complex regional pain syndrome (CRPS) type 1**
Can cause bilateral edema (Budapest Criteria); Diagnosis requires thorough history, physical exam, and tests to rule out other conditions.
- **Degenerative spondylolisthesis**
Leading to spinal stenosis. Inflammation and immune cascade post-spinal stenosis can resulting in CRPS symptoms (edema).

Discussion

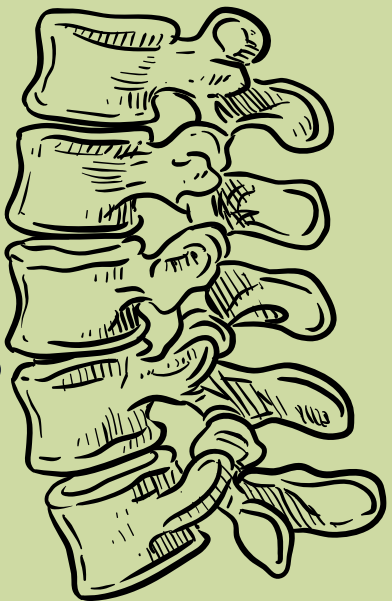


Peripheral edema

Often Linked to Cardiovascular, Liver, Renal Diseases, and Other Causes Like Pain, Drug Use, Lymphedema, Pregnancy, Low Activity Levels, and Mobility Limitation.

Degenerative spondylolisthesis

- Condition Involving Vertebral Slippage.
- Etiology: Disc Degeneration, Arthritis Of The Facet Joints, and Ligament And Muscle Dysfunction.
- More Common In Women (Ratio 6:1), Prevalence Of 25% ♀ Versus 19.1% ♂
- Predominantly occurs at The L4-L5 Level.
- Leading to Spinal Stenosis. Symptoms Include Low Back Pain and Radiating



^{Pain}Bilateral leg edema

May be associated with spinal conditions, but its direct connection between spinal stenosis is unclear.

Discussion

Pathophysiology of CRPS

- It involves abnormal tissue response, increased nervous system sensitization, inflammation, and autonomic dysregulation. Spinal stenosis exacerbation may trigger CRPS symptoms.
- Nociceptive C-fibres begin to produce inflammatory neuropeptides, activating mast cells and releasing chemical mediators that cause acute symptoms such as oedema, skin redness, warmth.

Treatment

- Fixed The Causa of Spinal Stenosis

Taylor SS, Noor N, Urits I, Paladini A, Sadhu MS, Gibb C, et al. Complex Regional Pain Syndrome: A Comprehensive Review. Vol. 10, Pain and Therapy. 2021.

Zale C, Hansen J, Ryan P. Complex Regional Pain Syndrome: Current Practice Management and Referral Trends in a Closed Healthcare System. Osteology. 2020;1(1).

Zangrandi A, Allen Demers F, Schneider C. Complex Regional Pain Syndrome. A Comprehensive Review on Neuroplastic Changes Supporting the Use of Non-invasive Neurostimulation in Clinical Settings. Vol. 2, Frontiers in Pain Research. 2021

Conclusion

Complex Regional Pain Syndrome (CPRS) Provoked by Spinal Stenosis with Bilateral Leg Edema, Resolved Through Decompression and Posterior Stabilization

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